

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD
CAMPUS

EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2026

TUTORIAL- 11

Date: 16-04-2026

AC Voltage Controller

Q1. A single-phase uni-directional AC voltage controller has input voltage of 120V RMS and 60Hz source and purely resistive load. The load resistance is 10Ω . The delay angles of the Thyristor T_1 and T_2 is $\alpha = 45^\circ$. Determine (a) output RMS voltage (b) the average input current (c) Input power factor (d) output power in watts

Q2. A single-phase bi-directional AC voltage controller has 120V and 60Hz source and purely resistive load. The load resistance is 15Ω . Determine (a) the delay angle required to deliver 500W to the R-load. (b) the source RMS current (c) RMS and average currents of the SCRs (d) input power factor (e) overall efficiency.

Q3. A single-phase bi-directional AC voltage controller has input voltage of 120V RMS and 60Hz source and RL load with load resistance is 20Ω and $L = 50\text{mH}$. The delay angles of the Thyristor T_1 and T_2 is $\alpha = 90^\circ$. Determine (a) an expression for load current for the positive half cycle at the steady state (b) for a conduction angle of 130° and 147W of heat power dissipation find (i) output RMS voltage (ii) the average currents of Thyristors (iii) RMS currents of thyristor (iv) Input power factor

Q4. An AC voltage ON-OFF controller has a resistive R load of $R = 10\Omega$ and supply voltage of $V_s = 120\text{V}$, 60Hz. The thyristor T_1 turns ON at $\alpha = 0^\circ$ and T_2 at $\alpha = 180^\circ$ for 25 cycles and remains OFF for 75 cycles. Determine (a) the RMS output voltage (b) the input power factor (c) the average and the RMS current of the thyristors

Q5. A single phase full-wave ac controller is connected to RL load. The input voltage is 230V, 50 Hz. If the firing angle 60° and extinction angle is 210° , calculate the rms load voltage. What is the conduction angle for each thyristor? Calculate the fundamental rms of the output voltage and its phase angle with respect to the source voltage?

Q6. A single-phase full-wave ac controller connected to a Heater (purely resistive, $R=25\Omega$) is supplied from a 230 V, 50 Hz power supply. If the firing angle is 90° , what is the output heat generated in kW? Calculate the input power factor, rms and average current flowing through each thyristor. If the same amount of heat needs to be generated using integral cycle control, what should the ratio, k?